

Are There Factors That Can Predict the Average Annual Cost of Fuel?

Predicting Annual Fuel Costs and Analyzing the Impact of Vehicle Characteristics on Fuel Efficiency

— Introduction

Context

Fuel expenses are one of the most significant and unpredictable long-term costs of vehicle ownership. Gas prices fluctuate, environmental concerns grow — predicting fuel cost has real-world urgency.

Research Questions

01. Are there factors that predict the average annual cost of fuel?

02. Is there an association between transmission type and combined fuel efficiency?

Why It Matters



Consumers

Fuel is one of the largest long-term costs of vehicle ownership. Understanding what drives cost helps buyers make smarter decisions.



Automakers

Identifying which features most impact efficiency guides engineering priorities and competitive positioning.



Lawmakers

Understanding vehicle characteristic–efficiency links informs emissions standards and incentive programs.

Data Overview

2025 EPA Fuel Economy Guide

868

Observations

23

Manufacturers

20

Vehicle Classes

TARGET VARIABLE

Annual Fuel Cost

PHYSICAL CHARACTERISTICS

Engine Displacement

Num of Cylinders

Drive Type

TECHNOLOGICAL INDICATORS

Transmission Type

Air Aspiration

Stop-Start System

Source: Official government standardized testing data.

 Cleaned: Dropped sparse variables (e.g., max_biodiesel_percent)

Key Summary Statistics

Metric	Value
Annual Fuel Cost (Mean)	\$2,641.65
Avg. Combined MPG	23.9 MPG
Avg. Num. Cylinders	5.3
Most Common Transmission	Semi-Automatic (44%)
Stop-Start Enabled	81.6%

**Averages based on 2025 EPA dataset subset.*

— Exploratory Data Analysis

Factors Affecting Fuel Cost



Fuel Efficiency

Strong Negative Association with Annual Fuel Cost.

Higher MPG = Lower Cost.

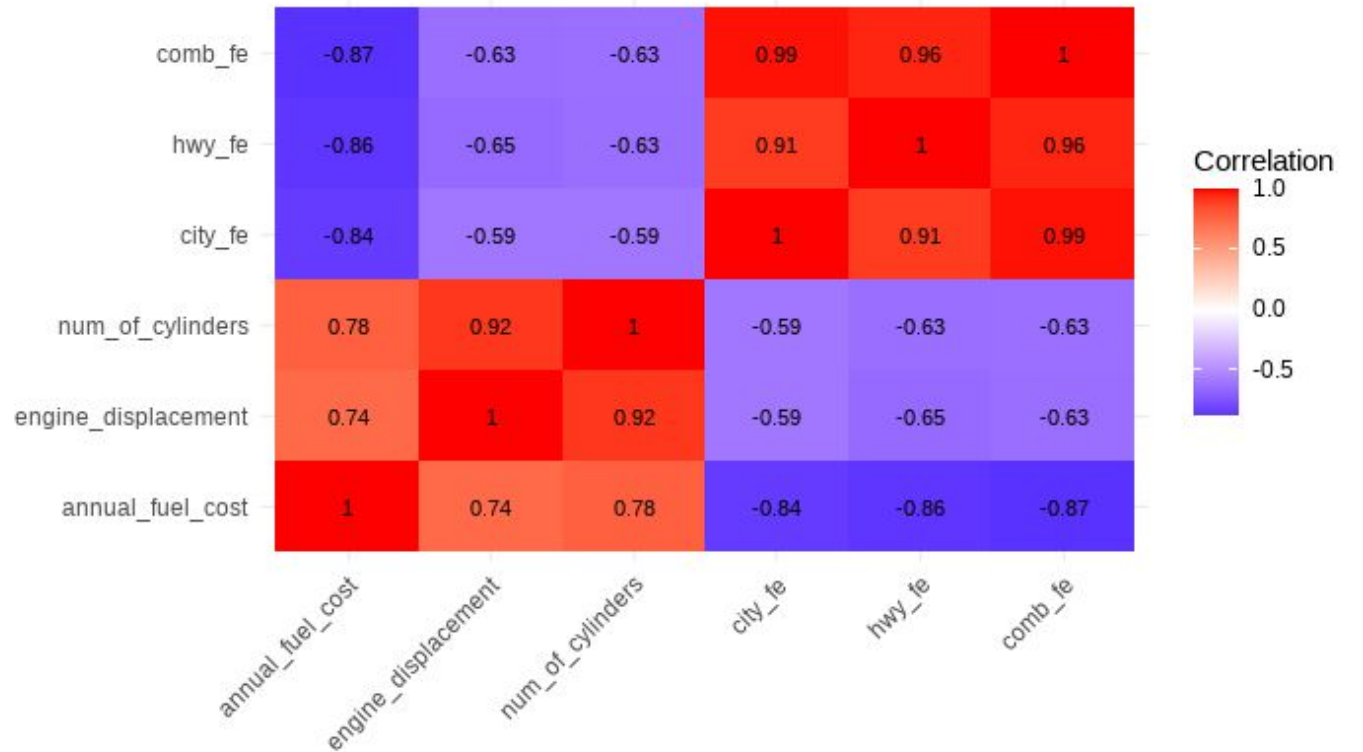


Engine & Cylinders

Strong Positive Association with Annual Fuel Cost.

Larger Engine = Higher Cost.

Correlation Matrix of Factors Affecting Fuel Cost



Multiple Linear Regression

Variance Explained

87.7% (Adjusted R²)

Key Predictors

- ✓ Combined Fuel Efficiency
- ✓ Number of Cylinders
- ✓ Drivetrain (AWD/4WD)
- ✓ Vehicle Class

Multiple Linear Regression Model Fit

Predicting annual fuel cost using combined FE, cylinders, drivetrain, and vehicle class

R ²	Adjusted R ²	Residual Std. Error	F-statistic	Model p-value	Num DF	Denom DF
0.877	0.876	276.219	556.996	0.0000	11.000	856.000

Multiple Linear Regression Coefficients

Factors associated with annual fuel cost.

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	3,447.09	96.02	35.90	0.0000
comb_fe	-73.13	2.00	-36.58	0.0000
num_of_cylinders	160.04	7.22	22.17	0.0000
drive_desc2-Wheel Drive, Rear	-26.07	36.80	-0.71	0.4789
drive_desc4-Wheel Drive	148.54	39.32	3.78	0.0002
drive_descAll Wheel Drive	125.39	29.35	4.27	0.0000
drive_descPart-time 4-Wheel Drive	12.07	50.63	0.24	0.8117
carline_class_descPassenger Car	100.33	53.89	1.86	0.0630
carline_class_descPerformance/Small	352.75	63.21	5.58	0.0000
carline_class_descPickup Truck	-154.72	60.89	-2.54	0.0112
carline_class_descSUV/Crossover	-29.56	52.68	-0.56	0.5748
carline_class_descVan/Minivan	-283.72	110.59	-2.57	0.0105

Blue cells indicate statistically significant predictors ($p < 0.05$)

Predictor of Annual Fuel Cost #1

Combined Fuel Efficiency

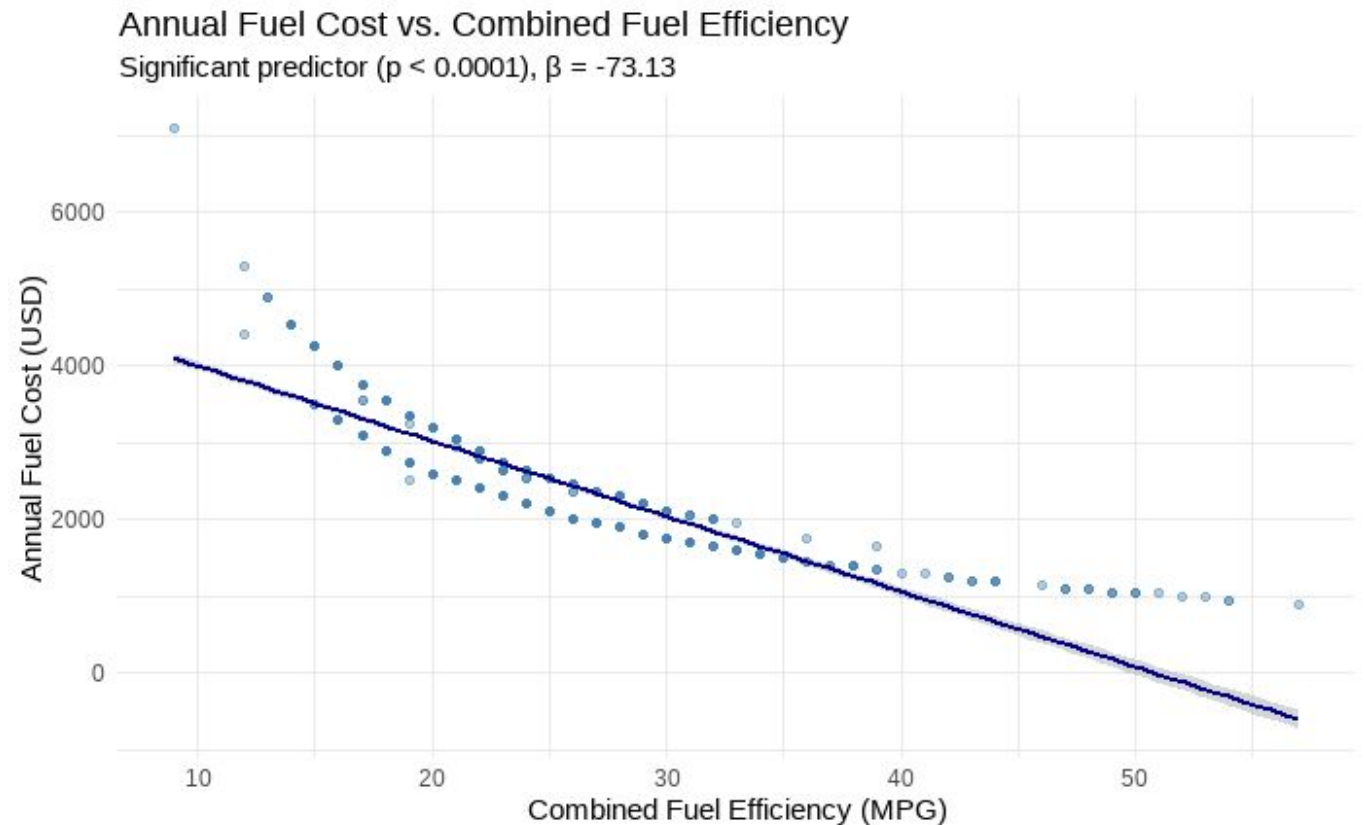
Stats

- $\beta = -73.13$ | $p < 0.0001$

Observations

- Every additional MPG reduces expected annual fuel cost by \$73.13, holding all else constant.
- The relationship is strong and negative — higher efficiency vehicles are dramatically cheaper to fuel.

Increased Efficiency = Less Expensive



Predictor of Annual Fuel Cost #2

Number of Cylinders

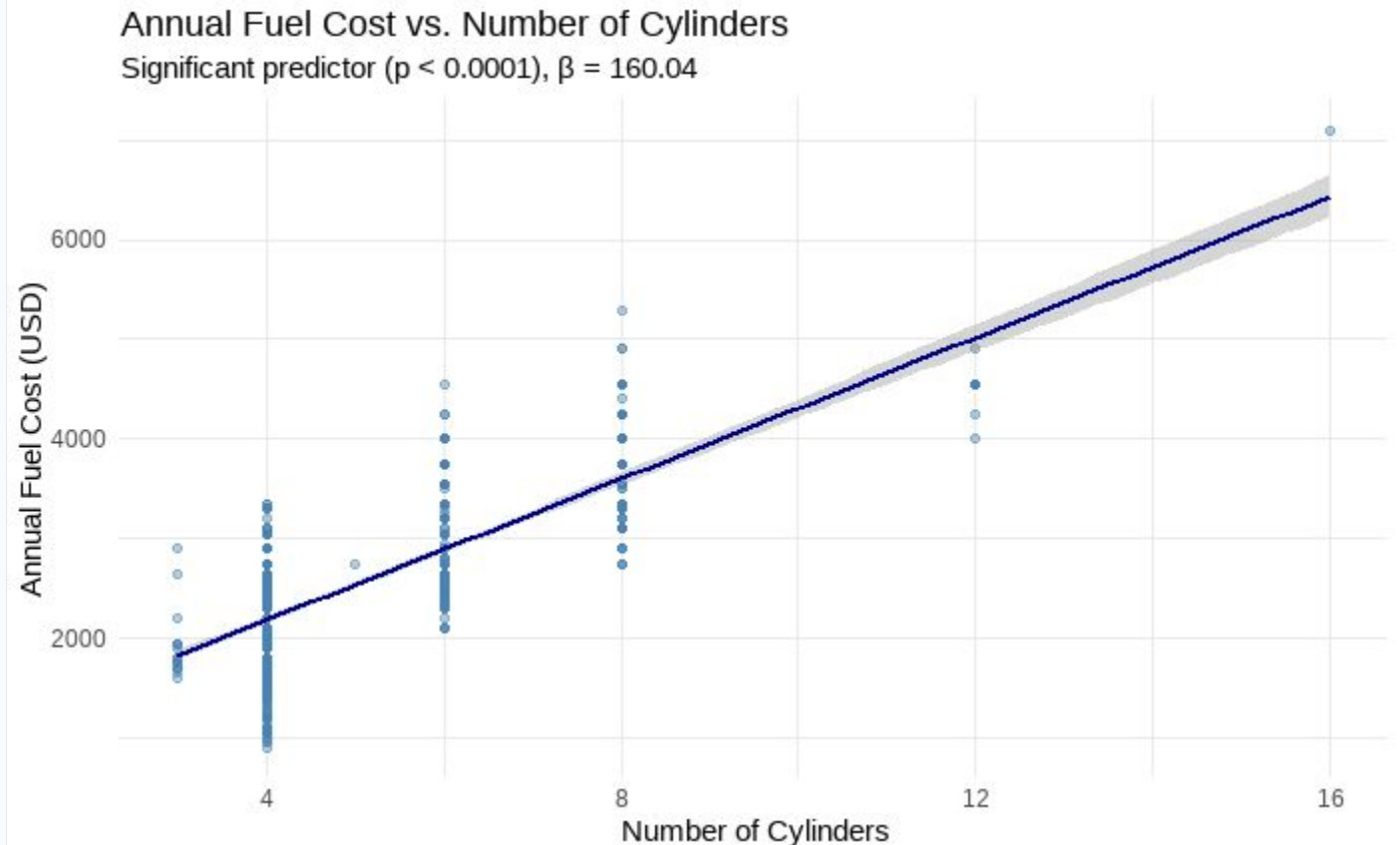
Stats

- $\beta = +160.04$ | $p < 0.0001$

Observations

- Each additional cylinder adds \$160 to expected annual fuel cost, controlling for efficiency and drivetrain.
- Cylinder count acts as a proxy for raw engine power demand.

More Cylinders = More Cost



Predictor of Annual Fuel Cost #3

Drivetrain Type

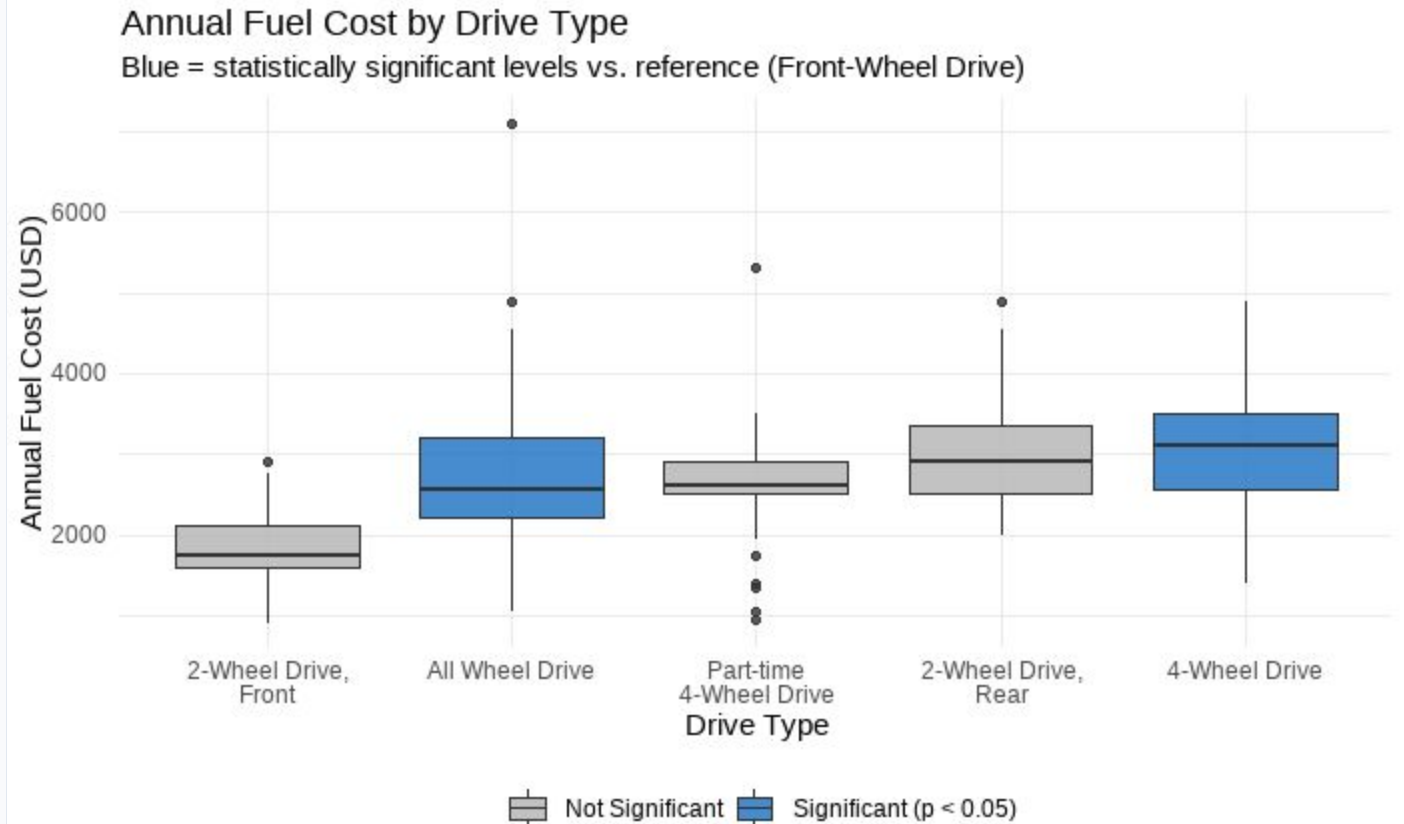
Stats

- AWD: $\beta = +125.39$
- 4WD: $\beta = +148.54$
 - (both $p < 0.05$)

Observations

- Relative to Front-Wheel Drive (reference), AWD adds ~\$125 and 4WD adds ~\$149 in annual fuel cost.
- Part-time 4WD and Rear-Wheel Drive are not statistically significant — only full-time systems show a clear penalty.

AWD or 4WD = Cost Premium



Predictor of Annual Fuel Cost #4

Vehicle Class

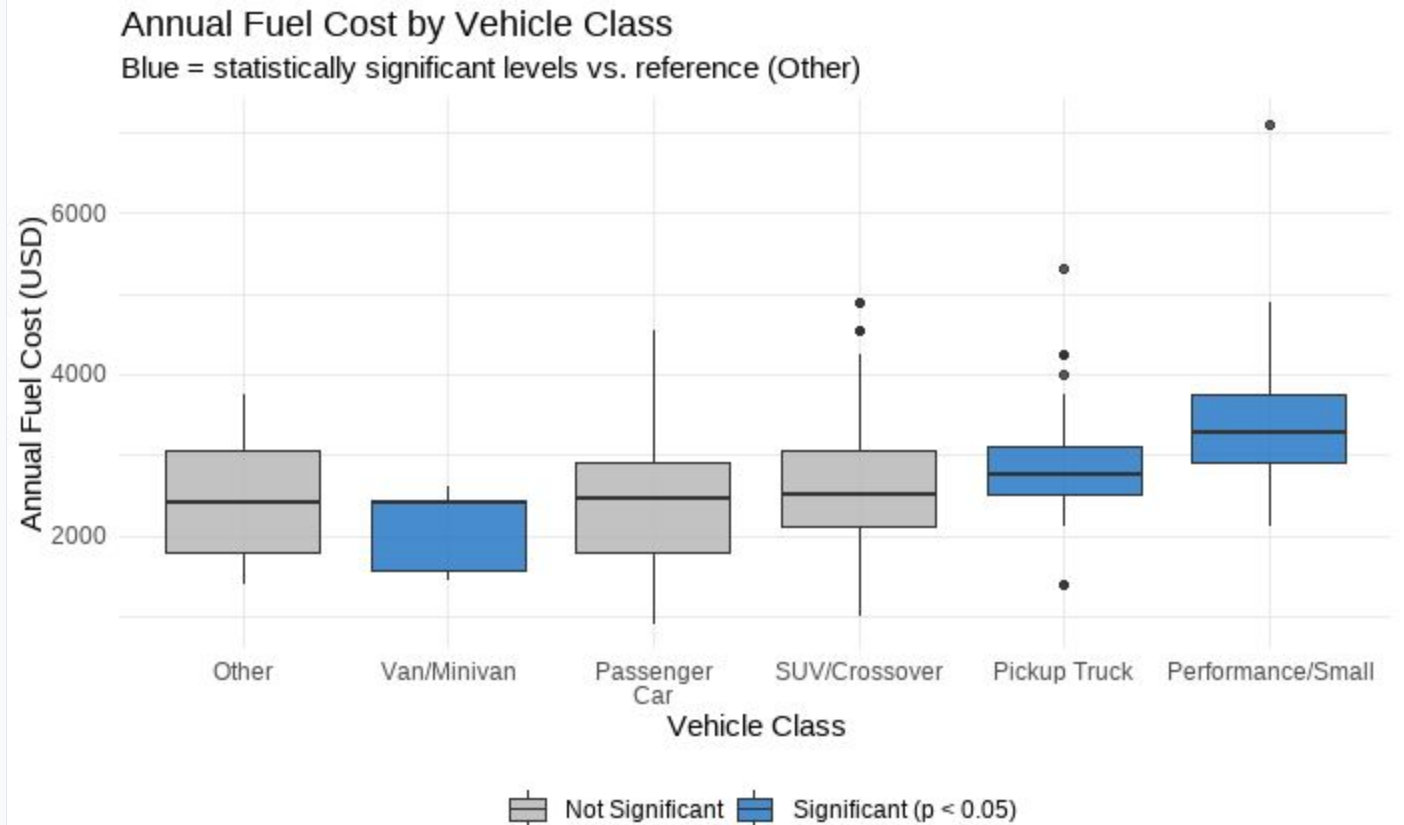
Stats

- Performance/Small: $\beta = +352.75$
- Pickup Truck: $\beta = -154.72$
- Van/Minivan: $\beta = -283.72$
 - (all $p < 0.05$)

Observations

- Performance/Small vehicles cost ~\$353 more per year than the "Other" reference class — likely driven by lower MPG despite smaller engines.
- Pickup Trucks and Vans are cheaper than expected after controlling for MPG and cylinders, suggesting these classes are more fuel-efficient relative to their engine size than their reputation implies.

Performance and Van Segments Show Distinct Cost Patterns



Model Performance

Variable Importance

Random Forest analysis confirms Combined Fuel Efficiency is the most valuable predictor, followed by Cylinders and Drivetrain.

Random Forest Variable Importance		
Increase in MSE and Node Purity when each variable is permuted		
Variable	% Increase in MSE	Increase in Node Purity
comb_fe	44.18	243,568,958.59
num_of_cylinders	28.87	121,270,364.57
drive_desc	30.32	58,407,117.53
carline_class_desc	25.54	20,436,035.65
Blue cells indicate the most important variable for that metric.		

Model Accuracy

Tree-based models significantly outperform linear models. XGBoost and Random Forest are the two that are significantly better than the rest of the models.

Final Model Scorecard				
Ranked by RMSE (lowest to highest)				
Model	RMSE ↓	MAE ↓	R-squared ↑	MAPE (%) ↓
XGBoost	183.4834	133.4891	0.9425	5.1639
Random Forest	184.8616	122.8410	0.9417	4.7618
Decision Tree	193.0693	140.1743	0.9366	5.5851
Elastic Net	266.4164	209.2647	0.8794	8.7503
OLS	266.5860	209.2748	0.8797	8.8100
Quantile Regression (Median)	278.9104	203.6260	0.8732	9.1003
Bayesian Regression	1964.7117	1897.1809	0.7866	74.0929

Conclusion

Key Takeaways

Fuel efficiency and engine size (cylinders) are the dominant drivers of annual fuel cost — not "surface" features. Drivetrain and vehicle class add meaningful penalties/premiums even after controlling for MPG and cylinders. Tree-based ensembles significantly outperform linear models, confirming non-linear relationships in the data.

Implications

Consumers (Who Want Affordability)

Prioritize MPG and cylinder count over marketing claims or badge prestige for long-term savings.

Manufacturers (Appealing to Affordability)

Improving combustion efficiency and right-sizing engines (lower cylinder counts where feasible) yield the highest impact on buyer fuel costs. AWD/4WD systems and performance-oriented classes carry measurable annual penalties.

Limitations

- High multicollinearity between engine displacement and cylinder count ($r = 0.92$) — we addressed this by retaining cylinders, but it underscores the importance of feature selection.
- EPA laboratory values do not reflect real-world variables (weight, aerodynamics, driving style) that influence actual fuel costs.
- Analysis is limited to the 2025 model year only.

Future Work

- Test interaction terms (e.g., cylinders $\times$ drivetrain) and explore additional regularization or stacking techniques.
- Incorporate multi-year EPA data to track efficiency trends over time and the effects of new powertrain technologies.



Thank You

Questions & Discussion

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